



MATERIAL SAFETY DATA SHEET

1. PRODUCT IDENTIFICATION

PRODUCT NAME: Choline Chloride 75% Liquid

SYNONYMS: Ethanaminium, 2-hydroxy-N,N,N-trimethyl-, chloride; biocolina; Hepacholine; Lipotril; Choline Chlorhydrate.

CAS No.: 67-48-1

Molecular Weight: 139.63

Chemical Formula: $\text{HOCH}_2\text{CH}_2\text{N}(\text{CH}_3)_3\text{Cl}$

Product Codes: 1582

2. COMPOSITION/INFORMATION ON INGREDIENTS

CAS#	Chemical Name	%	EINECS#
67-48-1	Choline Chloride	75	200-655-4

Hazard Symbols: None listed

Risk Phrases: None listed

3. HAZARDS IDENTIFICATION

Emergency Overview

Dry: Tan to off-white granule or powder with odor ranging from little to cereal. Hazards include dust explosion and dust irritation hazards. Absorbs moisture (hygroscopic) and may be slippery when spilled.

Aqueous: Colorless to light amber solution; slight amine (fish-like) odor; poses little or no immediate hazards.

Potential Health Effects

Eye: May cause eye irritation.

Skin: May cause skin irritation.



Ingestion:	May cause irritation of the digestive tract
Inhalation:	May cause respiratory tract irritation. The toxicological properties of this substance have not been fully investigated.
Chronic:	No information found

4.FIRST AID MEASURES

Eyes:	Flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Get medical aid.
Skin:	Get medical aid. Flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes.
Ingestion:	Get medical aid. Wash mouth out with water.
Inhalation:	Remove from exposure and move to fresh air immediately. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical aid.
Notes to Physician:	Treat symptomatically and

5.FIRE FIGHTING MEASURES

General Information:	As in any fire, wear a self-contained breathing apparatus in pressure-demand, MSHA/NIOSH (approved or equivalent), and full protective gear.
Extinguishing Media:	In case of fire, use water, dry chemical, chemical foam, or alcohol-resistant foam.

6.ACCIDENTAL RELEASE MEASURES

General Information:	Use proper personal protective equipment as indicated in Section 8.
Spills/Leaks:	Vacuum or sweep up material and place into a suitable disposal container

7.HANDLING AND STORAGE

Handling:	Avoid breathing dust, vapor, mist, or gas. Avoid contact with skin
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and eyes.

Storage: Store in a cool, dry place. Store in a tightly closed container

8.EXPOSURE CONTROLS/PERSONAL PROTECTION

Engineering Controls: Use adequate ventilation to keep airborne concentrations low.

Exposure Limits: CAS# 67-48-1:

Personal Protective Equipment

Eyes: Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

Skin: Wear appropriate protective gloves to prevent skin exposure.

Clothing: Wear appropriate protective clothing to prevent skin exposure.

Respirators: Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

9.PHYSICAL AND CHEMICAL PROPERTIES

Physical State:	Coherent liquid
Color:	Colorless
pH:	6.0-8.0
Vapor Pressure:	Negligible.
Viscosity:	Not available
Boiling Point:	Not available
Freezing/Melting Point:	303.00 - 305.00 deg C
Autoignition Temperature:	Not available
Flash Point:	Not available
Explosion Limits: Lower:	Not available
Explosion Limits: Upper:	Not available
Decomposition Temperature:	Not available
Solubility in water:	Soluble in water
Specific Gravity/Density:	Not available
Molecular Formula:	C5H14NOCl
Molecular Weight:	139.63



10. STABILITY AND REACTIVITY

Chemical Stability:	Stable under normal temperatures and pressures.
Conditions to Avoid:	Incompatible materials, exposure to moist air or water.
Incompatibilities with Other Materials:	Strong oxidizing agents, strong acids, strong bases.
Hazardous Decomposition Products:	Hydrogen chloride, nitrogen oxides, carbon monoxide, carbon dioxide, nitrogen.
Hazardous Polymerization:	Has not been reported.

11. TOXICOLOGICAL INFORMATION

RTECS#:	CAS# 67-48-1: KH2975000
LD50/LC50:	RTECS: CAS# 67-48-1: Oral, mouse: LD50 = 3900 mg/kg; Oral, rat: LD50 = 3400 mg/kg;
Carcinogenicity:	CHOLINE CHLORIDE, - Not listed as a carcinogen by ACGIH, IARC, NTP, or CA Prop 65.
Other:	See actual entry in RTECS for complete information

12. ECOLOGICAL INFORMATION

Not available

13. DISPOSAL CONSIDERATIONS

Dispose of in a manner consistent with federal, state, and local regulations

14. TRANSPORT INFORMATION

The product should be transported in a well-closed container to avoid sunlight and dampness, keep in a cool condition, this product is safe and not inflammable and not regulated by IMO.

15. REGULATORY INFORMATION



JUJIA BIOTECH
巨佳生物科技

SHANDONG JUJIA BIOTECH CO., LTD

Address: Jianghe No.2 Road, Zhanhua County, Shandong, China
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Health&Safety Reporting List

None of the chemicals are on the Health&Safety Reporting List.

European/International Regulations

European Labeling in Accordance with EC Directives

Hazard Symbols: Not available

Risk Phrases:

Safety Phrases:

S 24/25 Avoid contact with skin and eyes.

WGK (Water Danger/Protection)

CAS# 67-48-1: 1

Canada

CAS# 67-48-1 is listed on Canada's DSL List

US Federal

TSCA

CAS# 67-48-1 is listed on the TSCA Inventory.

16. OTHER INFORMATION

The information contained herein is based on data available to us and is believed to be correct, but we assume no responsibility for injury from the use of the product described herein.
